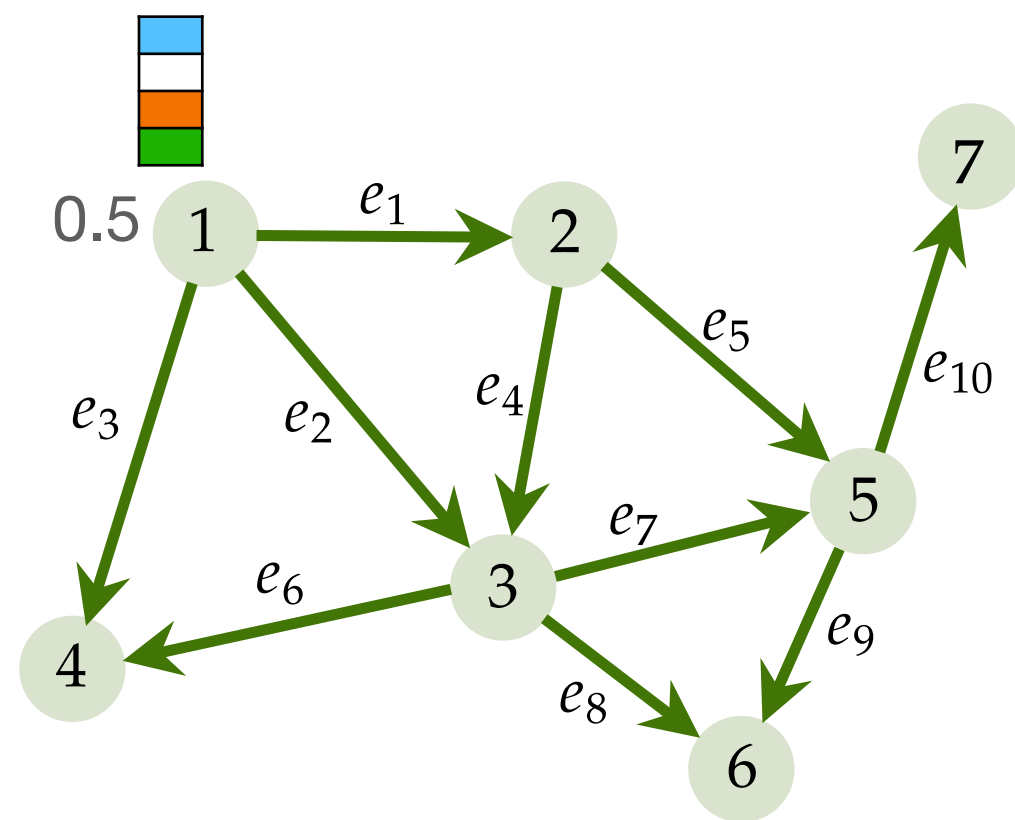


Data on simplices

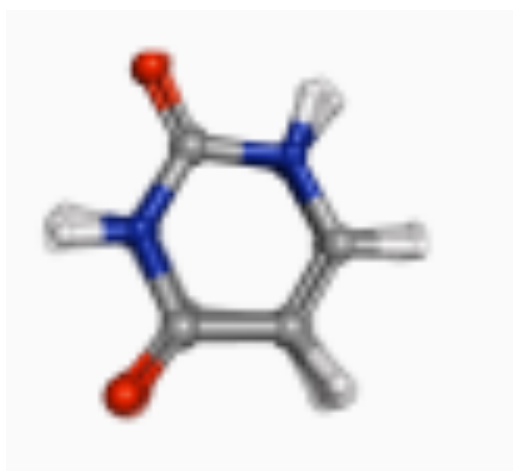
Signals on nodes, edges, triangles, ...



Node function

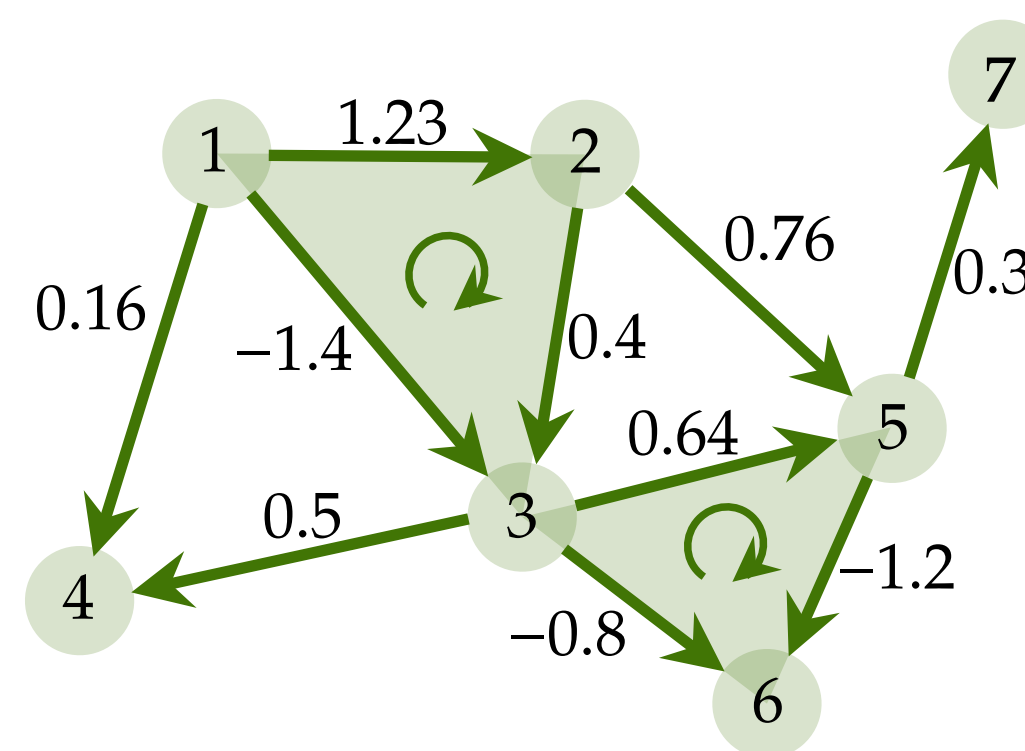
$$f_0 : V \rightarrow \mathbb{R}$$

$$\mathbf{f}_0 = (f_0(1), \dots, f_0(N_0))^T$$



Atom coordinates

$$f_0 : V \rightarrow \mathbb{R}^3$$



Edge function

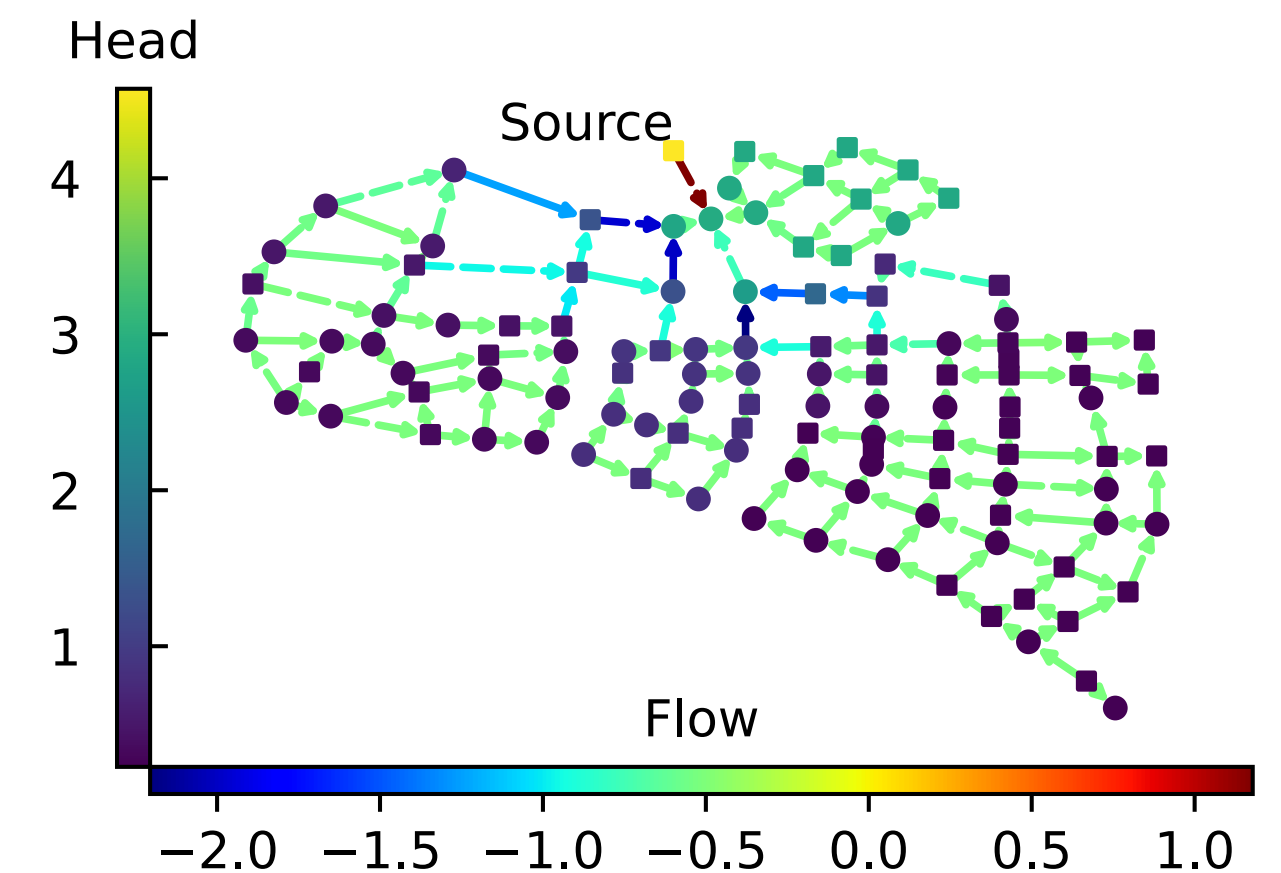
$$f_1 : E \rightarrow \mathbb{R}$$

$$\mathbf{f}_1 = (f_1(e_1), \dots, f_1(e_{N_1}))^T$$

- Alternating property

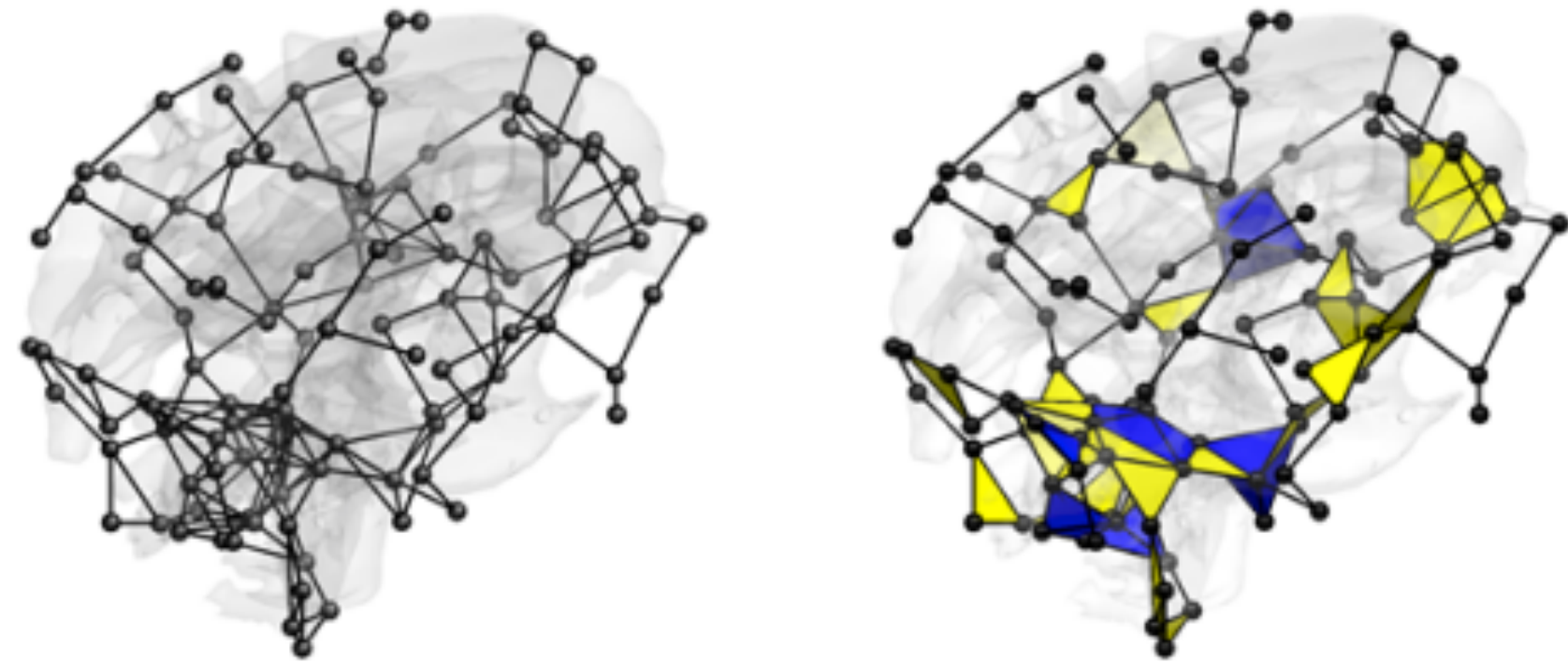
- Magnitude and sign

- Flow-type data (natural)
 - traffic/water flow, **information flow** in gene regulation...
- Edge-based vector field discretisation (computational)
- ...
- Representation learning
 - High-dim edge features
- Geometric algebra transformer 2023



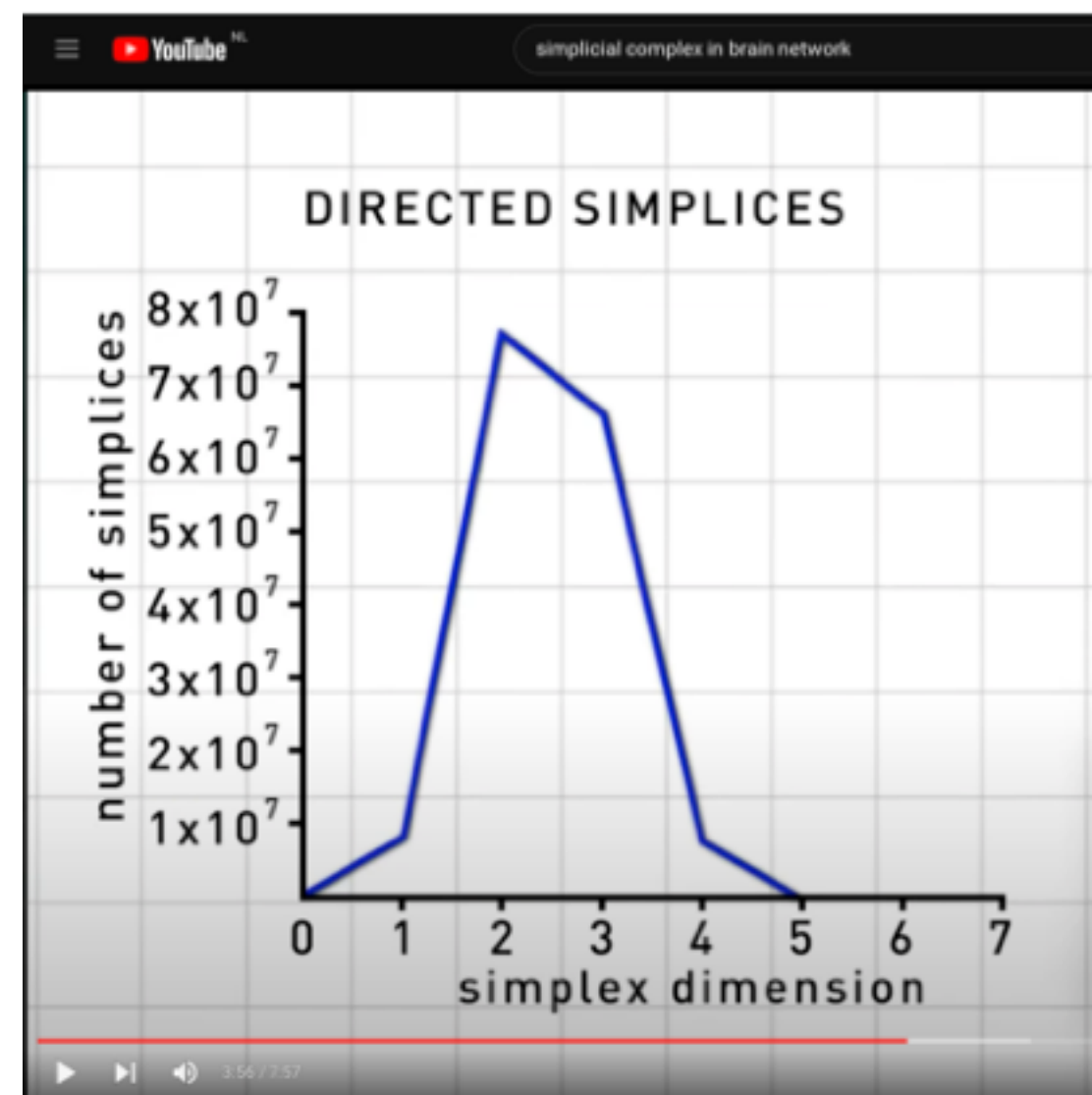
Water flows (Yang et al. 2023)

Simplicial complexes and Data in real world

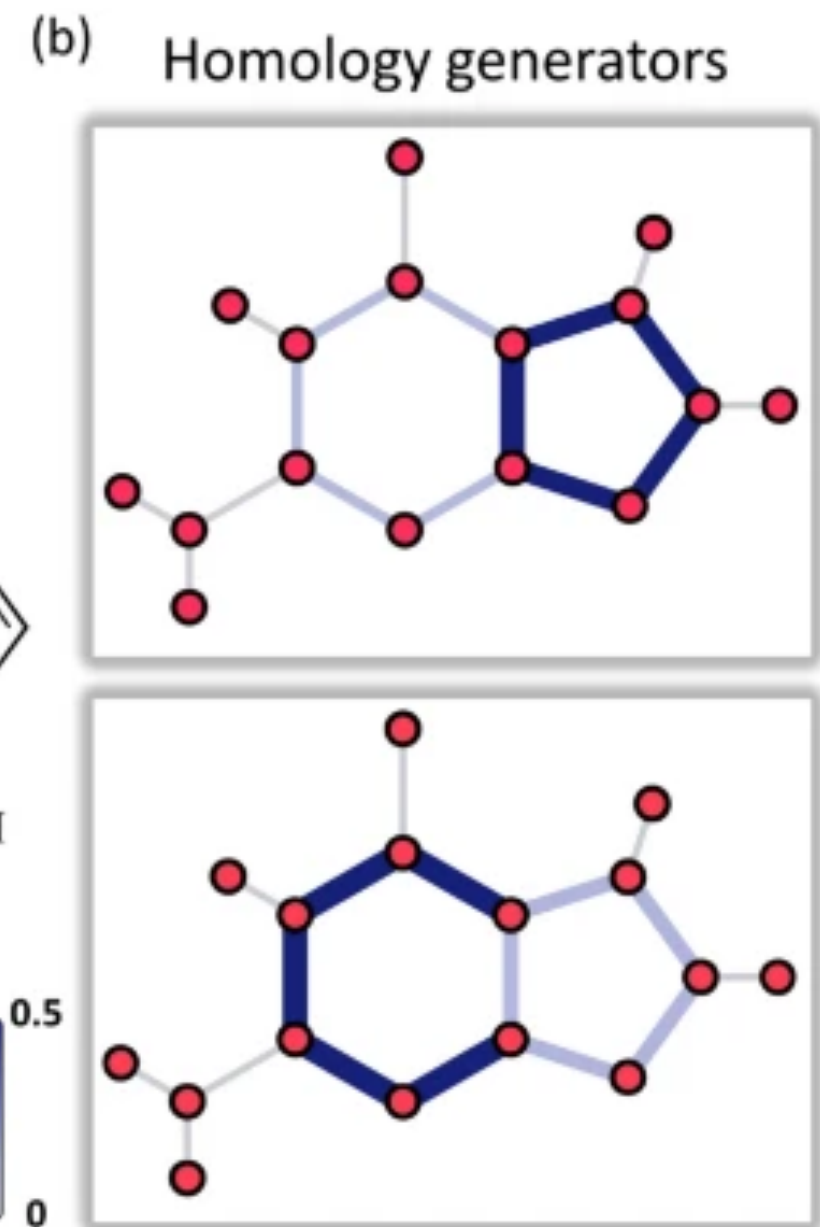
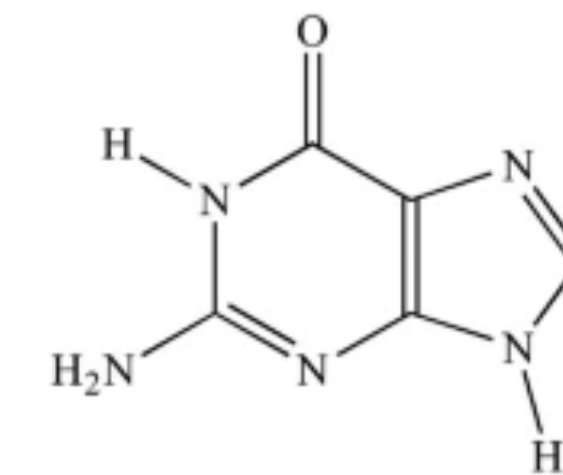
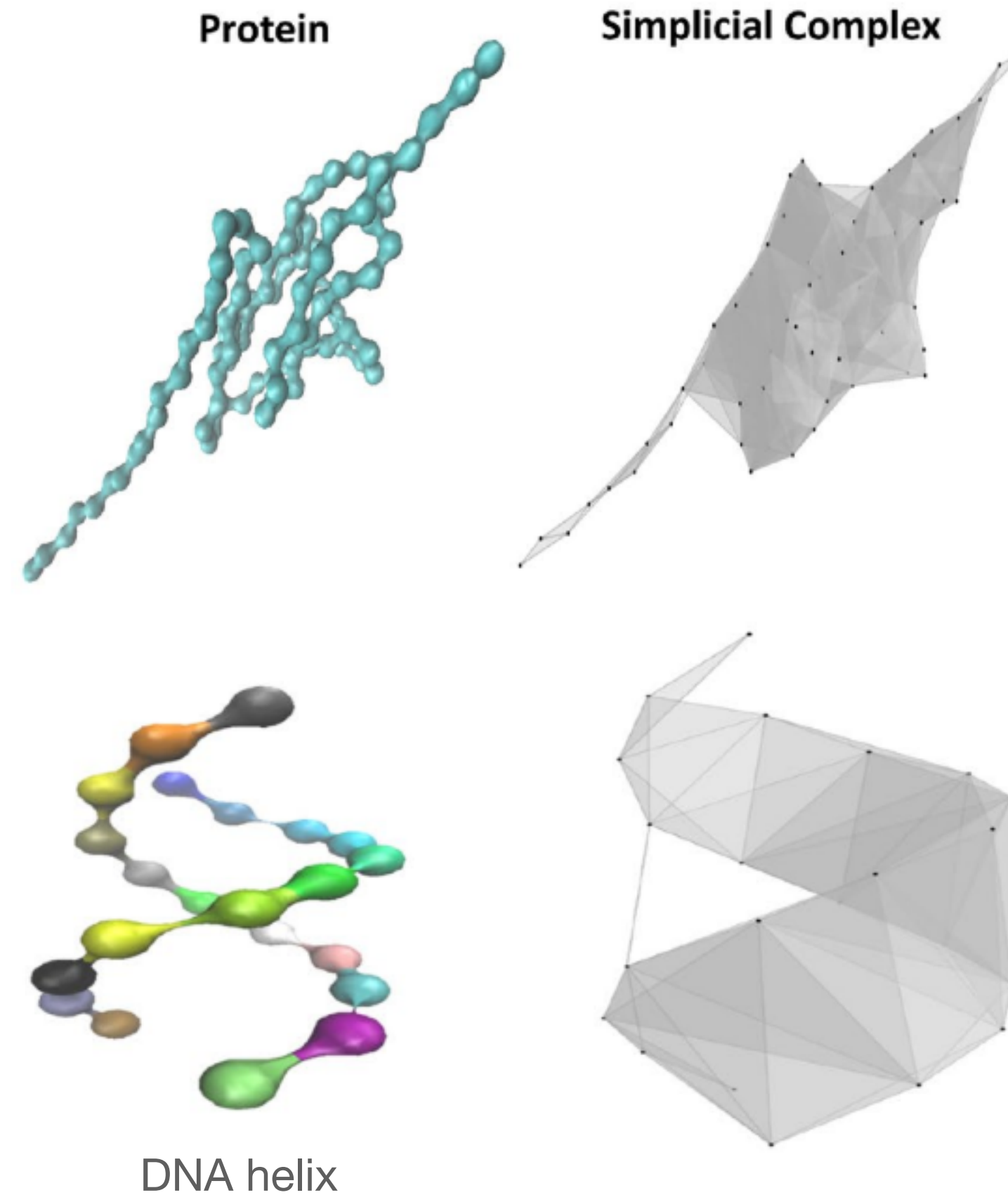


Neuroscience (Anand et al. 2023):

1. Firing of neurons
2. Activation of multiple brain regions
3. Triadic interactions in Schizophrenia (2025)



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biomolecular data (Wei et al. 2022)

biomolecular data (Wei et al. 2022)